



Computationally efficient deep learning models for diabetic retinopathy detection: a systematic literature review

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Abstract

Diabetic retinopathy, often resulting from conditions like diabetes and hypertension, is a leading cause of blindness globally. With diabetes affecting millions worldwide and anticipated to rise significantly, early detection becomes paramount. The survey scrutinizes existing literature, revealing a noticeable absence of consideration for computational complexity aspects in deep learning models. Notably, most researchers concentrate on employing deep learning models, and there is a lack of comprehensive surveys on the role of vision transformers in enhancing the efficiency of these models for DR detection. This study stands out by presenting a systematic review, exclusively considering 84 papers published in reputable academic journals to ensure a focus on mature research. The distinctive feature of this Systematic Literature Review (SLR) lies in its thorough investigation of computationally efficient approaches and models for DR detection. It sheds light on the incorporation of vision transformers into deep learning models, highlighting their significant contribution to improving accuracy. Moreover, the research outlines clear objectives related to the identified problem, giving rise to specific research questions. Following an assessment of relevant literature, data is extracted from digital archives. Additionally, in light of the results obtained from this SLR, a taxonomy for the detection of diabetic retinopathy has been presented. The study also highlights key research challenges and proposes potential avenues for further investigation in the field of detecting diabetic retinopathy.

Keywords Diabetic retinopathy · Deep learning · Computational complexity · Vision transformers

1 Introduction

Artificial intelligence (AI) has revolutionized a range of fields by incorporating human-like capabilities such as learning, reasoning, and perception into software systems. This technological progression has empowered computers to perform tasks traditionally handled by humans. Boosted by advancements in computing power, the availability of extensive

datasets, and the development of cutting-edge AI algorithms, AI applications are now widespread. Notable applications include finger vein recognition (Bilal et al. 2021), diabetic retinopathy detection (Bilal et al. 2024a, b, 2023, 2022, 2021), RNA Engineering (Yu et al. 2024; Feng et al. 2024), cancer detection (Bilal et al. 2024a, b, 2022a, b), biomathematical challenges (Bilal et al. 2021; Bilal and Sun 2020), and smart agriculture (Bilal et al. 2023).

The human eye stands as the paramount sensory organ responsible for detecting light and transmitting signals via the optic nerve to the brain. This process culminates in the formation of images, enabling the profound gift of vision, along with the ability to distinguish colors and perceive depth (Yadav and Singh 2019). Retinopathy, a condition that impacts the human eye's retina, typically results from various underlying diseases such as diabetes, hypertension, chronic kidney disease, and heart ailments (Fisher et al. 2016).

Diabetic retinopathy (DR) is a consequence of diabetes that causes vision damage and ultimately results in blindness if early diagnosis and treatment are delayed (Bhandari et al. 2023). Common indications of diabetic retinopathy encompass visual blurriness, the presence of floaters and flashes, as well as vision impairment or loss (Akram et al. 2013). For the vast number of diabetic patients, diabetic retinopathy (DR) has emerged as a major global medical problem and is currently the main contributor to blindness in people of working age (Thomas et al. 2019), (Raman et al. 2016; Bilal et al. 2021). Diabetes already affected 537 million people worldwide, according to a report conducted by the International Diabetic Federation in 2021 (Aschner et al. 2021). Globally, it is anticipated that the prevalence of diabetes will increase to 10.2% (578 million) and 10.9% (700 million) by 2030 and 2045 respectively (Saeedi et al. 2019). Additionally, it is predicted that the prevalence of impaired glucose tolerance will increase to 8.0% (454 million) and 8.6% (548 million) by the same time period. Around 75% of these individuals live in below middle-income nations, where almost 33% diabetic patients have DR (Wong and Sabanayagam 2020; Choo et al. 2021).

Hence, early detection of diabetic retinopathy is a pivotal objective, and the development of automated detection methods is imperative to alleviate the strain on the healthcare system caused by manual diagnosis (Bhandari et al. 2023; Dastane 2020). In this context, substantial research endeavors have been dedicated to the automatic detection of diabetic retinopathy, employing a combination of Machine Learning (ML) and image processing techniques (Priya and Aruna 2013). Both fields find applications in various medical contexts, including disease prediction, health risk analysis, and a range of other healthcare-related purposes.

After a thorough examination of the previous survey papers as shown in Table 1, it's evident that only a limited number of these papers delve into machine-learning methodologies for the detection of diabetic retinopathy. The majority of researchers have reviewed papers published up to the year 2021, with only a limited number of surveys including papers from 2022 in their examinations. Few researchers primarily focus on the segmentation techniques employed for diabetic retinopathy in their publications. However, the majority of the researchers place their emphasis on utilizing deep learning models for diabetic retinopathy detection. To the best of our knowledge, none of the past review papers have explored the computational complexity aspects associated with various deep learning models for diabetic retinopathy detection. There is also a notable absence of a comprehensive survey examining how vision transformers in deep learning models have contributed to improving the efficiency of models for DR detection. In this SLR, those papers are examined that are published in reputable academic journals, excluding workshop publications. This approach was

Table 1 Comparison with existing work

Review paper	Focus of review Paper	Review approach	Year	Quality assessment score	Explored review perspectives		Target repositories
					Transformer based model	Computationally efficient models	
Das et al. (2022)	ML and DL based approaches	Informal	2022	x	x	x	Not Given
Nadeem et al. (2022)	DL based approaches for DR detection and segmentation	Systematic	2022	x	x	x	4
Lalithadevi and Krishnaveni (2022)	DL based approached and image processing techniques for DR segmentation and detection	Systematic	2022	x	x	x	20
Tajudin et al. (2022)	CNN based approaches for DR detection	Informal	2022	x	x	x	Not Given
Bhandari et al. (2023)	DL and computational techniques for DR detection	Informal	2023	x	x	x	Not Given
Dayana and Emmanuel (2023)	DL and metaheuristic optimization approached for DR detection	Informal	2023	x	x	x	Not Given
Vij and Arora (2023)	DL based approaches for DR Detection	Systematic	2023	x	x	x	Not Given
Uppamma et al. (2023)	DL and Image processing techniques for DR detection	Informal	2023	x	x	x	Not Given
Radha and Karuna (2023)	ML and DL based Segmentation approached for DR	Informal	2023	x	x	x	Not Given
This SLR	DL based approached for DR detection and segmentation	Systematic	2023	10	✓	✓	7

chosen as it allows us to focus on more established and mature research, thus strengthening the rationale for our survey.

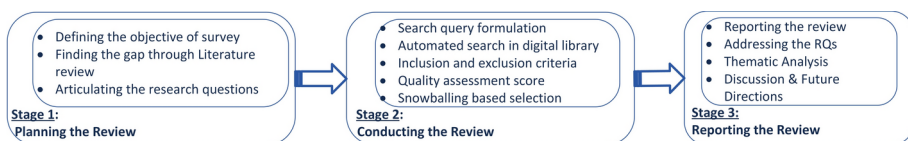
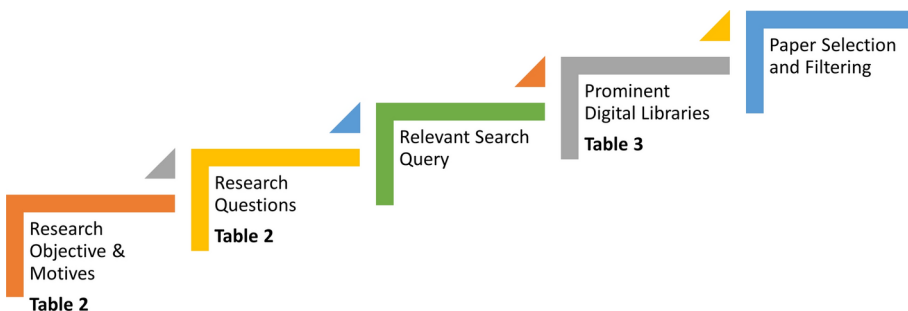
This SLR stands out by offering a thorough investigation of computationally efficient approaches and models. It includes an extensive study of how incorporating vision transformers into deep learning models can significantly improve the accuracy of diabetic retinopathy detection. This SLR will assist researchers in developing computationally efficient DL models for real-time Diabetic Retinopathy (DR) detection.

The major contributions of this research work are as follows:

1. A comprehensive, systematic, and in-depth review of computationally efficient deep learning models for diabetic retinopathy detection is performed.
2. The impact of using a vision transformer is analyzed to enhance the accuracy of deep learning models for diabetic retinopathy detection.
3. A detailed analysis and comparison of 13 publicly available datasets are provided.

Table 3 Query string for extracting a list of relevant SLRs

(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN OR Convolutional neural network) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization OR Computational) AND (Detection OR Segmentation OR Identification OR Classification) AND (Review OR Systematic Review OR Survey OR SLR)

**Fig. 1** Research Strategy**Fig. 2** Search Strategy

- A thematic analysis of the chosen studies is performed to provide a more clearer and comprehensive overview of the current state-of-the-art research for diabetic retinopathy detection.
- The immediate next steps and two potential research problems are proposed for future exploration in the field of diabetic retinopathy. The rest of the paper is defined as follows: The literature review is described in Section 2, Research methodology is explained in Section 3, Future recommendations are given in Section 4 and the Conclusion of this SLR is written in Section 5.

2 Literature review

The query string and relevant keyword groups used to extract the list of relevant SLRs are given in Table 3 and 2.

In their comprehensive research detailed in the paper by Das et al. (2022), a thorough exploration is conducted on the topics of Diabetic Retinopathy (DR), DR lesions, structural characteristics, challenges related to identification, and the sequential phases of incidence and progression. The study effectively utilizes DR images to create a chronological

Table 2 keyword groups for extracting a list of relevant SLRs

Term	Group 1	Group 2	Group 3	Group 4	Group 5
T1	Diabetic retinopathy	Deep Learning	Computational complexity	Detection	Review
T2	DR	Deep neural network	Computation Time	Segmentation	Systematic review
T3		Vision transformer	Lightweight	Identification	Survey
T4		Machine Learning	Efficient	Classification	SLR
T5		CNN	Optimization		
T6		Convolutional Neural network	Computational		

narrative. Furthermore, the research involves a comparative analysis of various Deep Learning (DL) models, including hybrid Machine Learning-Deep Learning (ML-DL) models, Deep Convolutional Neural Networks (DCNN), Convolutional Neural Networks (CNN), Transfer Learning (TL) models, ensemble ML/DL models, as well as evolutionary and comprehensive learning algorithms. This comparative analysis sheds light on their respective performances, particularly in the context of early DR detection. However, this review paper lacks a systematic approach, with a notable absence of information regarding the computational complexities of the models under scrutiny. Additionally, the study didn't mention any research questions. Furthermore, it predominantly draws upon papers published prior to 2021, with only a few exceptions from that year. Notably, the study does not incorporate research papers utilizing vision transformers for diabetic retinopathy detection.

Another study conducted by Nadeem et al. (2022) was carried out in a systematic manner, following the PRISMA approach. To identify the most pertinent published papers, the initial search query in this research paper was simple, containing only two keywords: "Diabetic retinopathy" and "deep learning." The authors of this study performed a thorough evaluation of different deep learning models, structures, and platforms, examining them with regard to their accuracy, the datasets they used, and the kinds of image modalities they applied. However, there is a scarcity of studies that have explored the challenges and consequences associated with deploying deep learning models in clinical environments. In this study, only two papers from the year 2022 were included, while the remaining papers featured in the study were published prior to 2022.

A comprehensive and thorough examination of automated diagnostic techniques for the detection of DR and related eye conditions was conducted by Lalithadevi and Krishnaveni (2022). This study explored various aspects, including the causes of DR, the availability of publicly accessible datasets, image preprocessing, the segmentation of different DR lesions, feature enhancement, an array of deep learning models, and the persistent challenges in ongoing research. Remarkably, among the papers reviewed in this study, only a limited number, approximately 6 to 7, were published in 2021, while the majority were from earlier years. Notably, this research does not delve into the computational complexities of the

discussed models. Additionally, this research falls short in providing guidance on which models are well-suited for real-time DR detection.

Tajudin et al. in their paper (Tajudin et al. 2022) provided an overview of 19 research papers published between 2017 and 2022 that utilize CNNs for the detection of five distinct severity levels of Diabetic Retinopathy (DR). The evaluation criteria employed to compare these CNN models encompass accuracy, the area under the receiver operating characteristic curve, specificity, and sensitivity. This review focused only on 19 papers, with two of them published in 2022, while the remaining are from 2017 to 2019 timeframe. The review process does not follow a systematic approach, and the research questions guiding these papers are not explicitly outlined. Furthermore, the scope of this study centers solely on CNN models in the context of DR detection.

A thorough analysis was carried out in another research on various approaches to detect DR, including detection methods and methods for selecting relevant features (Bhandari et al. 2023). The investigation covered a spectrum of computational techniques, such as fuzzy logic, particle swarm optimization, neural networks, and genetic algorithms. Additionally, the study delved into the use of combinations of these computational methods to select relevant features for diabetic retinopathy detection. However, it's worth noting that this review lacked a systematic approach. Additionally, the discussion of publicly available datasets for diabetic retinopathy detection was relatively limited in this research.

Dayana and Emmanuel (2023) in their research conducted an extensive and comprehensive analysis of the utilization of deep learning methods in the context of DR detection and grading based on fundus images. This review provided an in-depth exploration of the available fundus image datasets and the preprocessing techniques employed for diabetic retinopathy classification. Furthermore, this study systematically examined and discussed the categorization of diabetic retinopathy, employing transfer learning, ensemble learning, and metaheuristic optimization algorithms. However, the review lacks a systematic evaluation and does not include any research questions. Additionally, the computational complexity of the models is not incorporated into the study.

A systematic review conducted by Vij and Arora (2023) provided an extensive and in-depth analysis of the development of deep learning (DL) methods for the early-stage detection of diabetic retinopathy in retinal fundus images (RFIs). However, this study considered a review of papers related to diabetic retinopathy detection published up until 2021, and the computational complexity of models is also not discussed.

Another study (Uppamma et al. 2023) provided significant contributions in three key aspects. Firstly, it provided a comprehensive foundation on diabetic retinopathy (DR) as a medical condition, along with a thorough exploration of conventional diagnostic methods. Secondly, it offered an extensive survey of various imaging technologies and how deep learning is applied within the domain of DR. Thirdly, practical applications and real-world situations concerning DR detection are explained. However, this study did not follow a systematic approach, and there is an absence of well-defined research questions. Additionally, the study did not include lightweight models for the detection of DR. Similarly, the authors examined various machine learning and deep learning methods for blood vessel segmentation, along with an overview of publicly accessible datasets related to this field in another study (Radha and Karuna 2023). However, this review is not conducted in a systematic manner. Instead, it is specifically concentrated on segmentation techniques and does not

cover deep learning models for the automated detection of diabetic retinopathy. A list of available survey and review papers gathered through a search query is given in Table 3.

However, as compared to the previous reviews described above, this systematic literature review (SLR) stands out as this focuses exclusively on publication channels pertaining to Diabetic retinopathy. It takes a deep dive into the computational efficiency of models and identifies approaches that integrate vision transformers for the detection of Diabetic retinopathy. Additionally, this SLR approach is comprehensive and systematic. This study selects papers based on strict criteria described in Sect. 3.

3 Research methodology

The review process is structured into three key stages, as outlined in our research protocol: planning, execution, and data assessment. The search protocol is established after finalizing our research questions, which play a crucial role in guiding the retrieval of relevant review data and ensuring the impartial selection of studies.

3.1 Review plan

Figures 1 and 2 illustrate the methodology that outlines the research process for establishing the classification scheme, identifying pertinent publications, and defining the criteria for mapping publications. We have employed a comprehensive search strategy to capture all relevant DR data. The approach to study selection is systematically designed to ensure an unbiased process. Throughout this review, a well-structured procedure is followed that encompasses:

1. Defining research objectives
 2. Formulating research questions
 3. Curating a list of search repositories
 4. Meticulously selecting studies
 5. Thoroughly screening and evaluating results
 6. Extracting essential data
 7. Summarizing findings
 8. Concluding the review
- Creating research questions (RQs) and defining their objectives is a crucial step in conducting a comprehensive investigation of this study. The RQs and their objectives and motives are given in Table 4.

3.2 Review conduct

The review process has been structured into four distinct steps. In the first step, we conducted searches for pertinent primary studies from widely-used digital libraries. The second step involved the selection of studies, employing pre-defined inclusion/exclusion criteria. To enhance the quality of our review, the third step involved the quality assessment criteria. Finally, in the fourth step, we carried out backward snowballing to identify and extract significant candidate papers.

Table 4 Research question and their objective and motives

RQ	RQ statements	RQ objective and motives
RQ1	What are relevant publication channels for DR research?	To identify: · Reputable platforms for publishing scholarly works related to DR. · Research work produced from January 2021 to September 2023. · Conduct scientometric analysis using metadata, encompassing research categories, methodologies, and verification techniques
RQ2	What are the existing approaches and techniques for the automatic detection of DR?	· To gain insights into the various methods and frameworks used for detecting DR. · To comprehend the distinctions in models and methodologies employed in the context of ML and DL for this purpose
RQ3	What is the impact of using vision transformers in enhancing accuracy in DR detection models?	To grasp the incorporation of vision transformers into deep learning models for the detection of DR
RQ4	What are the various approaches and methodologies of deep learning that are computationally efficient for DR detection?	To identify the computationally efficient primary methods employed for detecting DR that include: · Lightweight · Computational Complexity
RQ5	What are the key performance metrics and benchmark datasets commonly used to evaluate the effectiveness and efficiency of deep neural networks for DR detection?	To identify datasets that are accessible and commonly employed evaluation metrics for the detection of DR

3.3 Automated search in digital libraries

A thorough and systematic research effort has been undertaken to sift through irrelevant studies and gather relevant information. To achieve this, a combination of automatic and manual search methods was employed when investigating the search terms. Throughout this process, numerous digital libraries are explored, opting to focus on repositories that are widely used in systematic literature surveys worldwide. Table 5 represents the digital venues as the primary sources for our automated search are selected, aiming to encompass nearly all pertinent research areas.

3.4 Search query

(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)

Table 5 Electronic search engine and digital libraries. DL represents digital library and SE represents search engine

Electronic database	URL	Type	Accessed date
Google Scholar	https://scholar.google.com/	SE	20 October 2023
WileyOnlineLibrary	https://onlinelibrary.wiley.com/	DL	20 October 2023
HEC Digital Library	http://www.digitallibrary.edu.pk/	DL	20 October 2023
ACM Digital Library	http://dl.acm.org	DL	20 October 2023
Pub Med	https://pubmed.ncbi.nlm.nih.gov/	DL	20 October 2023
IEEE Xplore	http://ieeexplore.ieee.org	DL	20 October 2023
ScienceDirect	https://www.sciencedirect.com	DL	20 October 2023

Table 6 keyword groups

Term	Group 1	Group 2	Group 3	Group 4
T1	Diabetic retinopathy	Deep learning	Computational complexity	Detection
T2	DR	Deep neural network	Computation Time	Segmentation
T3		Vision transformer	Lightweight	Identification
T4		Machine learning	Efficient	Classification
T5		CNN	Optimization	

Possible arrangements of search string can be seen in Fig. 3 while keyword groups and relative search query are given in Table 6 and in section 3.4. Moreover, the search strategy for digital libraries is given in Table 7.

3.5 Inclusion and exclusion criteria

3.5.1 Inclusion criteria

1. The article should have "Diabetic Retinopathy" or "Detection of DR" in its title.
2. The article must be published in a reputable journal or top-tier conference.
3. The publication date of an article should not be prior to 2019.
4. The article uses a dataset that is publicly available.
5. The article addresses one of the research questions.

3.5.2 Exclusion criteria

Articles are excluded from further study on the basis of the following conditions given below:

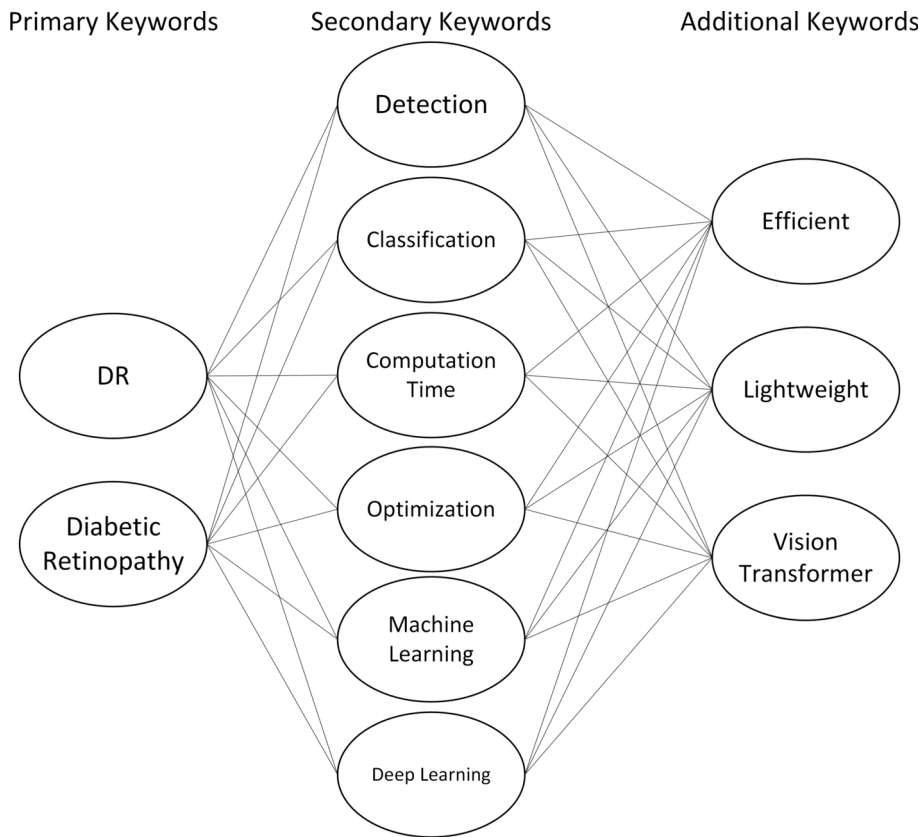


Fig. 3 Keywords used to develop search string

1. Exclude the articles that do not contain either “Diabetic Retinopathy” or “Detection of DR” in their titles.
2. Exclude articles that are not published in a reputable journal or top-tier conference.
3. The publication date of an article is prior to 2019.
4. The article uses a dataset that is not publicly available.
5. The article doesn’t address one of the research questions. Table 8 provides an overview of the noteworthy outcomes from the primary search, filtering, and inspection phases across seven digital libraries defined in section 3.3. I have just searched papers from Google Scholar, IEEE Xplore, Springer, Pub Med, ScienceDirect, and wiley Online library and added results in Table 8.

3.6 Quality assessment based selection

Following the application of inclusion and exclusion criteria, articles are subsequently narrowed down based on their quality assessment scores. The quality assessment process

Table 7 Search strategy for digital libraries

Digital library	Search query	Applied filter
Google Scholar	(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)	January 2021 to September 2023
Wiley Online Library	“(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)”	January 2021 to September 2023
ACM Digital Library	((All: Diabetic retinopathy) OR (All: DR)) AND ((All: Deep Learning) OR (All: Deep neural Network) OR (All: Vision Transformer) OR (All: Machine Learning) OR (All: CNN) AND ((All: Computational Complexity) OR (All: Computational Time) OR (All: Lightweight) OR (All: Efficient) OR (All: optimization)) AND ((All: Detection) OR (All: Segmentation) OR (All: Identification) OR (All: Classification))	January 2021 to September 2023
Pub Med	(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)	January 2021 to September 2023
IEEE Xplore	(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)	January 2021 to September 2023
Springer	(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)	January 2021 to September 2023
ScienceDirect	(Diabetic retinopathy OR DR) AND (Deep Learning OR Deep neural Network OR Vision Transformer OR Machine Learning OR CNN) AND (Computational Complexity OR Computational Time OR Lightweight OR Efficient OR optimization) AND (Detection OR Segmentation OR Identification OR Classification)	January 2021 to September 2023

Table 8 Selection phase and filtering

Phase	Selection	Selection criteria	Google scholar	Science direct	IEEE Xplore	Springer link	Wiley online library	PubMed	Total
1	Search	Keywords	29500	16596	493	4868	2162	876	54495
2	Filtering	Since 2021	16500	7291	278	3737	589	595	28990
3	Filtering	Abstract	1000	1430	36	430	250	190	3336
4	Filtering	Inspection	29	8	11	22	3	11	85

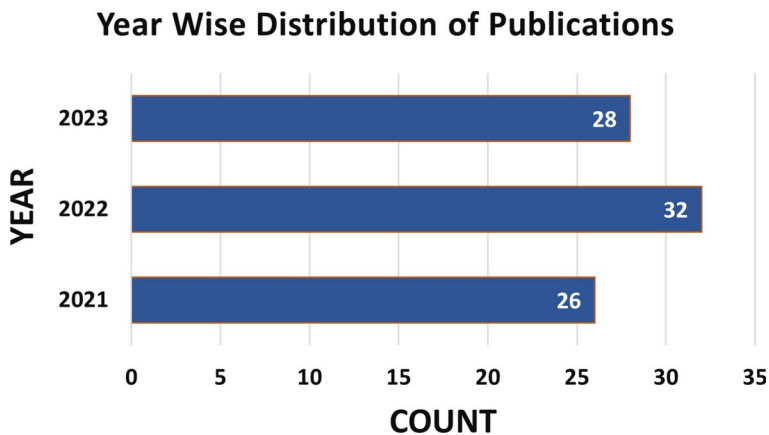


Fig. 4 Year-wise distribution of publications

stands as a pivotal step in conducting a systematic literature review. The total score is ten (10) and the score is calculated based on the following conditions given below:

1. Assigned a score of two (2) to articles published in journals and a score of one (1) to those published in top conferences.
2. Assigned a score of one (1) to the article if the dataset used in the article is publicly available otherwise assigned a zero (0) score.
3. Assigned a score of three (3) to the article if a vision transformer is used with a deep learning model, a score of two (2) if any deep learning model is used, and a score of one (1) if machine learning model is used.
4. Assigned a score of two (2) if it uses a lightweight model that is computationally efficient, otherwise assigned a score of one (1).
5. Assigned a score of one (1) if future work is given in the article, otherwise assigned a zero (0).
6. Assigned a score of one (1) if the results of the proposed model are compared with other state-of-the-art models, otherwise assigned a zero (0) score.

The quality assessment score of each selected study is given in Table 9. Only those papers are selected for further study whose quality assessment score is 5 or greater than 5.

3.7 Paper selection through snowballing

Following a quality assessment, a backward snowballing technique is used by reviewing the reference lists of each of the finalized studies to identify relevant papers. Only those papers are selected which fulfilled the defined criteria of inclusion and exclusion and their significance is determined based on an initial review of their abstracts and subsequent examination of the full content. After a comprehensive review of the selected papers, an additional three studies (Raiaan et al. 2023; Venkaiahpalaswamy et al. 2023), and Gu et al. (2023) are added to our survey.

Table 9 Quality assessment score

Sr#	References	Is the article published in a journal?	Is the used dataset publicly available?	Is the Vision transformer used?	Is the model light-weight?	Is future work given?	Are the results compared with other models?	Total score
1	(Khan et al. 2023)	2	1	3	2	1	1	10
2	(Luo et al. 2023)	2	1	2	1	0	1	7
3	(Ali et al. 2023)	2	1	2	1	0	1	7
4	(Raiaan et al. 2023)	2	1	2	2	1	1	9
5	(Jena et al. 2023)	2	1	2	1	0	1	7
6	(Sau and Bansal 2022)	2	1	2	1	1	1	8
7	(Bilal et al. 2022)	2	1	2	1	1	0	7
8	(Bilal et al. 2022)	2	1	2	1	0	1	7
9	(Özbay 2023)	2	1	2	1	0	1	7
10	(Aurangzeb et al. 2022)	2	1	2	2	0	1	8
11	(Dayana and Emmanuel 2022)	2	1	2	1	1	1	8
12	(Beham and Thanikaiselvan 2023)	2	1	2	1	1	0	7
13	(Krishnamoorthy et al. 2023)	2	1	2	1	1	1	8
14	(Elouaber et al. 2023)	2	1	2	1	0	1	7
15	(Rayavel and Murukesh 2022)	2	1	1	1	0	1	6
16	(Dutta et al. 2023)	2	1	3	1	1	1	9
17	(Bansode et al. 2023)	2	1	1	1	0	1	6
18	(Aziz et al. 2023)	2	1	2	1	0	1	7
19	(Pugal Priya et al. 2022)	2	1	1	1	0	1	6
20	(Raja Sarobin M and Panjanathan 2022)	2	1	2	1	1	1	8
21	(Hossain et al. 2023)	2	1	3	1	0	0	7
22	(Parthiban and Kamarasan 2023)	2	1	2	1	0	1	7
23	(Uppamma and Bhattacharya 2023)	2	1	2	1	0	1	7
24	(Nahiduzzaman et al. 2023)	2	1	1	2	0	1	7
25	(Dihin et al. 2022)	2	1	3	1	1	0	8
26	(Aujih et al. 2022)	2	1	2	2	0	1	8

Table 9 (continued)

Sr#	References	Is the article published in a journal?	Is the used dataset publicly available?	Is the Vision transformer used?	Is the model light-weight?	Is future work given?	Are the results compared with other models?	Total score
27	(Naik et al. 2023)	2	1	2	1	0	1	7
28	(Gunduru et al. 2022)	2	0	1	1	0	1	5
29	(Venkaiahppalaswamy et al. 2023)	2	1	2	1	1	1	8
30	(Abbood et al. 2022)	2	1	2	1	1	1	8
31	(Li et al. 2023)	2	1	3	1	0	1	8
32	(Kukkar et al. 2022)	2	1	2	2	0	1	8
33	(Ragab et al. 2022)	2	1	1	1	0	1	6
34	(Kshirsagar et al. 2022)	2	0	1	1	0	1	5
35	(Zhang et al. 2023)	2	1	3	2	0	1	9
36	(Jian et al. 2023)	2	1	1	2	1	1	8
37	(Wang et al. 2023)	2	1	3	1	1	1	9
38	(AbdelMaksoud et al. 2022)	2	1	2	1	1	1	8
39	(Gunasekaran et al. 2022)	2	1	2	1	0	0	6
40	(Dai et al. 2021)	2	1	2	1	0	0	6
41	(Abirami and Kavitha 2023)	2	1	2	2	0	1	8
42	(Murugappan et al. 2022)	2	1	2	1	0	1	7
43	(Al-Smadi et al. 2021)	2	1	2	1	0	1	7
44	(Wan et al. 2023)	2	1	3	2	0	1	9
45	(Khaparde et al. 2023)	2	1	2	1	0	1	7
46	(Mahmoud et al. 2021)	2	1	2	1	0	0	6
47	(Erciyas and Barışçi 2021)	2	1	2	1	1	1	8
48	(Shankar et al. 2021)	2	1	2	1	1	0	7
49	(Nguyen et al. 2021)	2	1	1	1	1	0	6
50	(Zhang et al. 2021)	2	1	2	2	0	1	8
51	(Elwin et al. 2022)	2	1	2	1	0	1	7
52	(Rajamani and Sasikala 2023)	2	1	1	1	1	1	7
53	(Narhari et al. 2021)	2	1	1	1	0	1	6

Table 9 (continued)

Sr#	References	Is the article published in a journal?	Is the used dataset publicly available?	Is the Vision transformer used?	Is the model light-weight?	Is future work given?	Are the results compared with other models?	Total score
54	(Nair and Muthuvel 2021)	5	1	2	1	0	1	5
55	(Saeed et al. 2021)	2	1	2	1	1	1	8
56	(Abdelmaksoud et al. 2021)	2	1	2	1	1	1	8
57	(Hassan et al. 2022)	2	1	2	1	0	1	7
58	(Bodapati et al. 2021)	2	1	2	1	1	0	7
59	(Khan et al. 2022)	2	0	2	1	1	0	6
60	(Vasireddi et al. 2022)	2	1	2	1	0	1	7
61	(Goel et al. 2021)	2	1	2	1	0	0	6
62	(Das et al. 2021)	2	1	2	1	0	1	7
63	(Dayana and Emmanuel 2022)	2	1	2	1	1	1	8
64	(Sungheetha et al. 2021)	2	1	1	1	1	0	6
65	(Ramasamy et al. 2021)	2	1	1	1	0	1	6
66	(Menaouer et al. 2022)	2	1	2	1	1	1	8
67	(Qureshi et al. 2021)	2	1	2	1	1	1	8
68	(Butt et al. 2022)	2	1	2	1	1	1	8
69	(Khan et al. 2021)	2	1	2	1	1	1	8
70	(Saini and Susan 2022)	2	1	2	1	1	1	8
71	(Nawaz et al. 2021)	2	1	2	1	1	1	8
72	(Elloumi et al. 2022)	2	0	2	2	1	0	7
73	(Elloumi et al. 2021)	2	1	2	2	1	1	9
74	(Kanmozhi et al. 2021)	2	1	2	2	0	1	8
75	(Nneji et al. 2022)	2	1	2	2	0	1	7
76	(Taranum and Rajashekar 2022)	2	1	2	1	1	0	7
77	(Palaniswamy and Vellingiri 2023)	2	1	2	1	0	1	7
78	(Liu et al. 2022)	2	1	2	1	1	1	8
79	(Al-Antary and Arafat 2021)	2	1	2	1	0	1	7
80	(Fan et al. 2021)	2	1	2	1	1	1	8

Table 9 (continued)

Sr#	References	Is the article published in a journal?	Is the used dataset publicly available?	Is the Vision transformer used?	Is the model light-weight?	Is future work given?	Are the results compared with other models?	Total score
81	(Vinayaki and Kalaiselvi 2022)	2	1	2	1	0	1	7
82	(Kadan and Subbian 2021)	2	1	2	1	0	1	7
83	(Gupta et al. 2022)	2	1	2	2	0	1	8
84	(Albahli et al. 2021)	2	1	2	1	1	1	8
85	(Modi and Kumar 2023)	2	1	2	1	0	1	7
86	(Gu et al. 2023)	2	1	3	1	1	1	9
87	(Bhardwaj et al. 2021)	2	1	2	1	1	1	8

3.8 Assessment of RQ1: what are relevant publication channels for DR research?

Year-wise distribution and geographic distribution for a total of 86 selected papers are given in Figs. 4 and 5 respectively.

The data extracted from each article includes the publication year, the name of the journal in which it was published, the dataset utilized within the article, the accuracy of the model presented in the article, and the model introduced as part of the methodology within the article. Table 10 and 11 shows the extracted information according to the defined criteria.

3.9 Assessment of RQ2: what are the existing approaches and techniques for the automatic detection of DR?

The evaluation of the existing approaches and techniques for the automatic detection of Diabetic Retinopathy (DR) reveals a diverse landscape of methodologies employed across various studies. Based on our examination of the chosen studies detailed in Table 11, it is evident that a variety of deep learning models have been introduced and preferred for the detection of Diabetic Retinopathy (DR). The predominant trend observed among researchers is the widespread utilization of deep learning models in addressing this objective. Khan et al. proposed DR-CCTNet, achieving 90.17% accuracy on datasets APTOS, Messidor2, IDRiD, DDR, and Kaggle Diabetic Retinopathy (Khan et al. 2023). Luo et al. (2023) explored a variety of deep learning models, including DenseNet, VGG, Inception Net, MobileNet V2, ResNext50, ShuffleNet, ViT, and ViG. A Hybrid Convolutional Neural Network, achieving an impressive 96.85% accuracy with the EyePACS dataset is also employed in another study (Ali et al. 2023). The RetNet-10 model demonstrated high accuracy (98.65%) in DR detection using datasets APTOS, Messidor2, and IDRiD introduced by Raiaan et al. (2023). Sau

Geographic Distribution of Publication

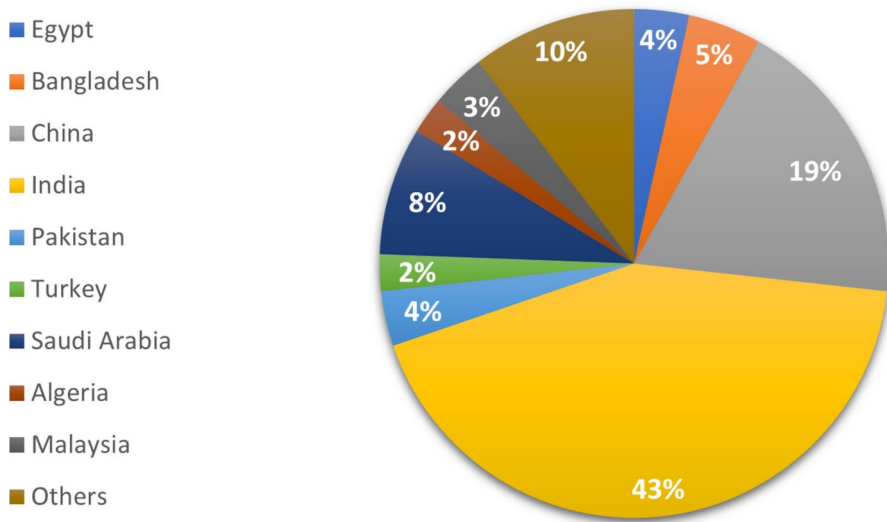


Fig. 5 Geographic distribution of publications

Table 10 Publication sources

References	Publication source	Year	Publication type	Count	Percentage
Khan et al. (2023)	Biomedicines	2023	Journal	1	1%
Luo et al. (2023)	CAAI Transactions on Intelligence Technology	2023	Journal	1	1%
Ali et al. (2023)	IEEE Journal of Translational Engineering in Health and Medicine	2023	Journal	1	1%
Raiaan et al. (2023), Aurangzeb et al. (2022), Abbood et al. (2022), Saeed et al. (2021), Abdelmaksoud et al. (2021), Khan et al. (2021), Al-Antary and Arafa (2021), Palaniswamy and Vellingiri (2023)	IEEE Access	2023, 2022, 2021	Journal	8	9%
Jena et al. (2023)	Big Data and Cognitive Computing	2023	Journal	1	1%
Parthiban and Kamarasan (2023), Qureshi et al. (2021), Gupta et al. (2022), Dayana and Emmanuel (2022), Sau and Bansal (2022)	Multimedia Tools and Applications	2022, 2023, 2021	Journal	5	6%
Elaouaber et al. (2023), Bansode et al. (2023), Bilal et al. (2022)	Computer Methods in Biomechanics and Biomedical Engineering: Imaging Visualization	2022, 2022	Journal	3	3%
Raja Sarobin M and Panjanathan (2022), Bilal et al. (2022)	Symmetry	2022	Journal	2	2%
Özbay (2023), Krishnamoorthy et al. (2023)	Artificial Intelligence Review	2023	Journal	2	2%
Beham and Thanikaiselvan (2023), Naik et al. (2023)	Soft Computing	2023	Journal	2	2%
Rayavel and Murukesh (2022)	The Imaging Science Journal	2022	Journal	1	1%
Dutta et al. (2023), Li et al. (2023)	Journal of Imaging	2023	Journal	2	2%
Aziz et al. (2023)	Scientific Reports	2023	Journal	1	1%
Pugal Priya et al. (2022)	International Journal for Numerical Methods in Biomedical Engineering	2022	Journal	1	1%
Uppamma and Bhattacharya (2023), Fan et al. (2021)	Electronics	2023, 2021	Journal	2	2%
Nahiduzzaman et al. (2023)	Expert Systems with Applications	2023	Journal	1	1%
Dihin et al. (2022)	International Journal of Innovative Computing	2022	Journal	1	1%
Aujih et al. (2022)	IEEE Transactions on Bio-medical Circuits and Systems	2022	Journal	1	1%
Ragab et al. (2022), Kshirsagar et al. (2022), Khan et al. (2022), Gundluru et al. (2022)	Computational Intelligence and Neuroscience	2022	Journal	4	5%

Table 10 (continued)

References	Publication source	Year	Publication type	Count	Percentage
Elwin et al. (2022), Das et al. (2021), Venkaiahppalawamy et al. (2023)	Biomedical Signal Processing and Control	2023, 2022, 2021	Journal	3	3%
Bhat and Anoop (2023)	International Journal of Image and Graphics	2023	Journal	1	1%
Kukkar et al. (2022)	IEEE Transactions on Computational Social Systems	2022	Journal	1	1%
Zhang et al. (2023)	Journal of Cancer Research and Clinical Oncology	2023	Journal	1	1%
Saini and Susan (2022), Jian et al. (2023)	Computers in Biology and Medicine	2023, 2022	Journal	2	2%
AbdelMaksoud et al. (2022)	Medical Biological Engineering Computing	2022	Journal	1	1%
Gunasekaran et al. (2022)	BioMed Research International	2022	Journal	1	1%
Dai et al. (2021)	Nature communications	2021	Journal	1	1%
Abirami and Kavitha (2023)	Signal, Image and Video Processing	2023	Journal	1	1%
Murugappan et al. (2022)	Measurement	2022	Journal	1	1%
Al-Smadi et al. (2021)	International Journal of Electrical and Computer Engineering	2021	Journal	1	1%
Wan et al. (2023)	Applied Sciences	2023	Journal	1	1%
Khaparde et al. (2023)	Sensing and Imaging	2023	Journal	1	1%
Mahmoud et al. (2021)	Personal and Ubiquitous Computing	2021	Journal	1	1%
Erciyas and Barışçı (2021)	Computational and Mathematical Methods in Medicine	2021	Journal	1	1%
Nawaz et al. (2021), Albahli et al. (2021), Shankar et al. (2021), Nguyen et al. (2021)	Computers, Materials Continua	2021	Journal	4	5%
Zhang et al. (2021), Rajamani and Sasikala (2023)	Informatica	2021, 2023	Journal	2	2%
Narhari et al. (2021)	Bio-Algorithms and Med-Systems	2021	Journal	1	1%
Nair and Muthuvel (2021)	Journal of Mechanics in Medicine and Biology	2021	WJournal	1	1%
Hassan et al. (2022)	Frontiers in Medicine	2022	Journal	1	1%
Kanimozhi et al. (2021), Bodapati et al. (2021)	Journal of Ambient Intelligence and Humanized Computing	2021	Journal	2	2%
Vasireddi et al. (2022)	Graefe's Archive for Clinical and Experimental Ophthalmology	2022	Journal	1	1%
Goel et al. (2021)	Mathematical Problems in Engineering	2021	Journal Limited	1	1%
Bhardwaj et al. (2021), Dayana and Emmanuel (2022)	Neural Computing and Applications	2022, 2021	Journal	2	2%
Suntheetha et al. (2021)	Journal of Trends in Computer Science and Smart Technology	2021	Journal	1	1%
Ramasamy et al. (2021)	PeerJ computer science	2021	Journal	1	1%
Menaouer et al. (2022)	SN Computer Science	2022	Journal	1	1%

Table 10 (continued)

References	Publication source	Year	Publication type	Count	Percentage
Nneji et al. (2022), Butt et al. (2022)	Diagnostics	2022	Journal	2	2%
Elloumi et al. (2022)	International Symposium on Intelligent Data Analysis	2022	Journal	2	2%
Taranum and Rajashekar (2022)	Traitement du Signal	2022	Journal	1	1%
Liu et al. (2022)	Plos one	2022	Journal	1	1%
Vinayaki and Kalaiselvi (2022)	Neural Processing Letters	2022	Journal	1	1%
Kadan and Subbian (2021)	International Journal of Imaging Systems and Technology	2021	Journal	1	1%
Modi and Kumar (2023)	Computers Industrial Engineering	2023	Journal	1	1%

et al. utilized MDNN, a Modified Deep Neural Network (MDNN) that achieved an accuracy of 95.20% using the IDRiD dataset (Sau and Bansal 2022). Bilal et al. (2022) in their study used U-Net, achieving accuracies of 96.60%, 93.95%, and 92.25% with EyePACS-1, Messidor-2, and DIARETDB0 datasets, respectively. Similarly, some other techniques and models e.g. Conv-ViT (Dutta et al. 2023), ADL CNN (Özbay 2023), lightweight DL model based on ColonSegNet (Aurangzeb et al. 2022), Chronological Tunicate Swarm Algorithm (CTSA) (Dayana and Emmanuel 2022), Inception V3 model (Beham and Thanikaiselvan 2023), RCNN-GAN (Krishnamoorthy et al. 2023), SegNet (Elaouaber et al. 2023), Standard Convolutional Neural Network (CNN), and many others have been introduced in the chosen papers for DR detection. Techniques like LSTM-RFO (Pugal Priya et al. 2022), PCNN (Parallel Convolutional Neural Network) (Nahiduzzaman et al. 2023), Shifted windows (Swin) Transformers (Dihin et al. 2022), EDR Net (Aujih et al. 2022), HemNet, Swin Transformer V2 (Li et al. 2023), Hybrid (RCNN-GAN), Tripple cascade convolutional neural network (Jian et al. 2023), and Autoregressive-Henry (Elwin et al. 2022) have also been used for DR detection.

3.10 Assessment of RQ3: what is the impact of using vision transformers in enhancing accuracy in DR detection models?

Vision transformers (ViTs) are emerging as a potent alternative to convolutional neural networks (CNNs) in diabetic retinopathy (DR) detection, addressing limitations inherent in the traditional CNN approach. While CNNs excel at extracting local spatial features from fundus images, they may lack a comprehensive global context, struggle with long-range dependencies, and impose high computational costs. ViTs overcome these challenges by dividing images into patches treated as tokens and leveraging attention mechanisms to capture both local and global information. This offers advantages such as enhanced contextual understanding, increased flexibility in modeling long-range dependencies, and improved scalability. Existing end-to-end DL models are proficient at processing either texture or shape-based information independently, limiting their ability to impart the necessary robustness for classifying various retinal diseases. Addressing this limitation, Dutta et al. (2023) introduced a fusion model named 'Conv-ViT' specifically designed for detecting retinal dis-

Table 11 Information extraction table

References	Publisher	Accuracy	Dataset	Methodology
Khan et al. (2023)	MDPI	90.17%	APTOS, Messidor2, IDRiD, DDR, and Kaggle Diabetic Retinopathy	DR-CCTNet
Luo et al. (2023)	Wiley Online Library	93.50%	Messidor and EyePACS	DenseNet, VGG, Inception Net, MobileNet V2, ResNext50, ShuffleNet, ViT, ViG
Ali et al. (2023)	IEEE	96.85%	EyePACS	Hybrid Convolutional Neural Network
Raiaan et al. (2023)	IEEE	98.65%	APTOS, Messidor2, and IDRiD	RetNet-10
Jena et al. (2023)	MDPI	98.6% and 91.9%	APTOS and MESSIDOR	Deep Learning
Sau and Bansal (2022)	Springer	95.20%	IDRiD dataset	MDNN (modified deep neural network)
Bilal et al. (2022)	Taylor Francis	96.60%, 93.95%, and 92.25%	EyePACS-1, Messidor-2, and DIARETDB0	U-Net
Bilal et al. (2022)	MDPI	97.92%	Messidor-2, EyePACS-1, and DIARETDB0	U Net
Özbay (2023)	Springer	99.66%	EyePacs	ADL CNN
Aurangzeb et al. (2022)	IEEE	96.59%, 98.9%, and 97.3%	DRIVE, CHASE_DB1, and STARE	lightweight DL model based on ColonSegNet
Dayana and Emmanuel (2022)	Springer	95.% and 95.48%	DIARETDB0 and DIARETDB1	optimized deep neural network with Chronological Tunicate Swarm Algorithm (CTSA)
Beham and Thani-kaiselvan (2023)	Springer	95%	DR Kaggle database	novel Inception V3 model
Krishnamoorthy et al. (2023)	Springer	99%	APTOS, Kaggle, and Messidor	Hybrid (RCNN-GAN)
Elaouaber et al. (2023)	Taylor Francis	97.70%, 97.98%, and 95.52%	DRIVE, HRF, and CHASE-DB1	U Net, SegNet, and Standard convolutional neural network (CNN)
Rayavel and Murukesh (2022)	Taylor Francis	96.95%	ORIGA, CHASE_DB1, STARE, and DRIVE	CNN, RNN, SVM, FKM and DA
Dutta et al. (2023)	MDPI	94.46, 92.37	Mendeley Dataset, OCTID Dataset	Conv-ViT
Bansode et al. (2023)	Taylor Francis	96%	'High Resolution Fundus (HRF), Messidor, and DIARETDB'	SS-JO-CN-LSTM

Table 11 (continued)

References	Publisher	Accuracy	Dataset	Methodology
Aziz et al. (2023)	Nature Publishing Group UK London	97.19%	DIARETDB1 and DIARETDB0	HemNet
Pugal Priya et al. (2022)	Wiley On-line Library	97.93%	MESSIDOR, STARE, and DRIVE	LSTM with deep LSTM-RFO
Raja Sarobin M and Panjanathan (2022)	MDPI	96.22%, 93.18% and 75.61%		CNN, hybrid CNN with ResNet, hybrid CNN with DenseNet 2.1
Hossain et al. (2023)		82.26%	Diabetic Retinopathy Dataset	Vision Transformer (ViT) model with Shifted Patch Tokenization (SPT) and Locality Self Attention (LSA)
Parthiban and Kamarasan (2023)	Springer	9.65%	MESSIDOR	intelligent coyote optimization algorithm with DL based DR detection and grading (ICOA-DLDRD) model
Uppamma and Bhat-tacharya (2023)	MDPI	94.20%	IndianDRImage Dataset (IDRID)	TaylorAVO-SqueezeNet framework
Nahiduz-zaman et al. (2023)	Elsevier	91.7% and 97.2%	Kaggle DR 2015 competition Dataset-1, APTOS 2019	PCNN (Parallel Convolutional Neural Network)
Dihin et al. (2022)		85.30%	APTOS 2019 Kaggle benchmark dataset	Shifted windows (Swin) Transformers
Aujih et al. (2022)	IEEE	92% and 95%	DRKaggle and Messidor-2	EDR Net
Naik et al. (2023)	Springer	96.64%	e Asia Pacific Tele-Ophthalmology Society (APTOS-2019)	CNN Based Model
Gundluru et al. (2022)	Hindawi	96.7%	Diabetic Retinopathy Debrecen dataset from the UCI ML library	Neural Network with PCA for feature extraction
Venkaiahppalaswamy et al. (2023)	Elsevier	94% and 94.83%	DIARETDB0 and DIARETDB1	AlexNet and GoogleNet as CNN based network models used for classifying images of diabetic retinopathy
Abbood et al. (2022)	IEEE	93.60%	EyePACS and MESSIDOR	Resnet50
Li et al. (2023)	MDPI	99.9% and 99.5%	OCT2017 and OCT C8	Swin Transformer V2
Kukkar et al. (2022)	IEEE	95.78%	MESSIDOR	diabetic retinopathy classification (DRC) system by hybridizing the DL model

Table 11 (continued)

References	Publisher	Accuracy	Dataset	Methodology
Ragab et al. (2022)	Hindawi	95%	Deep-Diabetic-Retinopathy-Image-Dataset-(DeepDRiD)	CNN Model
Kshirsagar et al. (2022)	Hindawi	95.90%		Neural Network
Zhang et al. (2023)	Springer	86.91%	APTOS2019 and CIFAR-10	SMiT
Jian et al. (2023)	Elsevier	92.08%	APTOS 2019 Blindness Detection dataset	Tripple cascade convolutional neural network
Wang et al. (2023)		86.04%	IDRiD	Transformer-based model named "VTA + HE"
Abdel-Maksoud et al. (2022)	Springer	91.2%	APTOS 2019, IDRiD, EyePACS, MESSIDOR	E-DenseNet
Gunasekaran et al. (2022)	Hindawi	-	MESSIDOR	DL
Dai et al. (2021)	Nature Publishing Group UK London	0.972 AUC	SIM, CNDACS, NDSP, EyePACS	Deep Learning
Abi-rami and Kavitha (2023)	Springer	98.50%	MESSIDOR and IDRiD	KMA-optimized ABiLSTM
Murugap-pan et al. (2022)	Elsevier	99.73 %	APTOS2019	DRNet
Al-Smadi et al. (2021)	IAES Institute of Advanced Engineering and Science	-	APTOS 2019	DL
Wan et al. (2023)	MDPI	93.67%	Diabetic Retinopathy 224x224 (2019 Data)	MobileViT-Plus
Khaparde et al. (2023)	Springer	96.60%	DIARETDB1 database	Attention-based swin U-net
Mahmoud et al. (2021)	Springer	96.62%	CHASE	hybrid inductive machine learning algorithm (HIMLA)
Erciyas and Barışçi (2021)	Hindawi Limited	99.1% and 100%	Kaggle and MESSIDOR	DL
Shankar et al. (2021)	Tech Science Press	99.37%	MESSIDOR	DL
Nguyen et al. (2021)	Tech Science Press	98.40%	Messidor DR	DL

Table 11 (continued)

References	Publisher	Accuracy	Dataset	Methodology
Zhang et al. (2021)	Institute of Informatics	97.55%	MESSIDOR and IDRiD	DL
Elwin et al. (2022)	Elsevier	91.40%	IDRID	Autoregressive-Henry Gas Sailfish Optimization (Ar-HGSO)-based deep learning
Rajamani and Sasikala (2023)	Informatica	-	Kaggle	CNN
Narhari et al. (2021)	De Gruyter	92%, 93% and 95%	HRF, Messidor, and DIARETDB	FR-CSA (Fitness Rat based Crow Search Algorithm)
Nair and Muthuvel (2021)	World Scientific	94.10%	DIARETDB0 and DIARETDB1	Deep Convolutional Neural Network
Saeed et al. (2021)	IEEE	9.73% and 99.88%	EyePACS and Messidor	
Abdelmaksoud et al. (2021)	IEEE	95.10%	DRIVE, HRF, STARE, CHASEDB1, DIARETDB0, DIARETDB1, MESSIDOR, E-ophtha	ML-CAD
Hassan et al. (2022)	Frontiers	99.38%	HRF, MESSIDOR-2, IDRiD	DL
Bodapati et al. (2021)	Springer	82.54%	APTOS-2019	DL
Khan et al. (2022)	Hindawi	96.68%	Private	DL
Vasireddi et al. (2022)	Springer	97.60%	MESSIDOR	DL
Goel et al. (2021)	Hindawi Limited	95.90%	IDRID	DL
Das et al. (2021)	Elsevier	98.70%	DIARETDB1	DL
Dayana and Emmanuel (2022)	Springer	96%	DIARETDB0 and DIARETDB1	DL
Sungheetha et al. (2021)		97%	EyePACS	ML
Ramasamy et al. (2021)	PeerJ Inc	97.05%	DIARETDB1 and Kaggle	ML
Menaouer et al. (2022)	Springer	90%	APTOS-2019	DL
Qureshi et al. (2021)	Springer	98%	EyePACS	ADL-CNN
Butt et al. (2022)	MDPI	97.80%	APTOS	DL

Table 11 (continued)

References	Publisher	Accuracy	Dataset	Methodology
Khan et al. (2021)	IEEE	83.8% AUC	EyePACS	VGG-NIN
Saini and Susan (2022)	Elsevier	-	Kaggle DR detection, IDRiD and DDR	DL
Nawaz et al. (2021)	Tech Science Press	96.60%	SD-OCT	DL
Elloumi et al. (2022)	Springer	95.91%	Private	DL
Elloumi et al. (2021)	SPIE	98.36%	DIARETDB1	DL
Kani-mozhi et al. (2021)	Springer	98.06%	DRIVE, STARE, DIARETDB1, MESSIDOR	DL
Nneji et al. (2022)	MDPI	98% and 98.5%	Kaggle and Messidor	DL
Taranum and Rajashekar (2022)		97%	IDRID	EWLSM-DL
	IEEE	98.97%	MESSIDOR	IoTDL-DRD
Palaniswamy and Vellingiri (2023)				
Liu et al. (2022)	Public Library of Science San Francisco, CA USA	96.83%	MTNet	MTNet
Al-Antary and Arafat (2021)	IEEE	84.60%	EyePACS and APTOS	MSA-Net
Fan et al. (2021)	MDPI	85.32%	APTOS 2019	DL
Vinayaki and Kalaiselvi (2022)	Springer	95.42%	DRIVE	MTRO
Kadan and Subbian (2021)	Wiley Online Library	-	DIARETDB1	DL
Gupta et al. (2022)	Springer	98.90%	APTOS-2019 and EYEPACS	ML
Albahli et al. (2021)	Tech Science Press	97.20%	Kaggle and APTOS-2019	Densenet-65 Based Faster-RCNN
Modi and Kumar (2023)	Elsevier	94.61%	RFMID	DL
Bhardwaj et al. (2021)	Springer	90.51%	IDRID	DL

Table 11 (continued)

References	Publisher	Accuracy	Dataset	Methodology
Gu et al. (2023)	Hindawi	90.51%	DDR and IDRID	ViT (Vision Transformer) and CSRA (class specific residual attention)

eases from optical coherence tomography (OCT) images. To address both texture and shape-based features comprehensively, the proposed Conv-ViT combines transfer learning-based CNN models, including Inception-V3 and ResNet-50, for processing texture information by calculating pixel correlations. Additionally, they incorporated a vision transformer model to process shape-based features by evaluating pixel correlations over longer distances. This hybrid approach effectively integrates these three models to facilitate shape-based texture feature learning during the classification of retinal diseases.

In another study, Wang et al. (2023) introduced a novel Transformer-based model, termed "VTA + HE," designed specifically for Diabetic Retinopathy (DR) multi-lesion segmentation tasks. This model enhances the original Vision Transformer architecture by integrating a spatial prior module, utilizing convolutional neural networks for image feature extraction. Additionally, the architecture incorporates a spatial feature injector and extractor to boost feature interaction. Their proposed model exhibits superior performance in segmentation on the IDRiD dataset. Hossain et al. (2023) implemented Vision Transformer (ViT) with Dual Shifted Patch tokenization (SPT) and Locality self-attention (LSA) that addresses the inherent location inductive bias (LIB), resulting in enhanced accuracy on relatively smaller datasets compared to traditional Vision Transformer models. To tackle class imbalance, they used StyleGAN2, a generative model, to generate synthetic images for underrepresented classes. A comparative analysis is also conducted by implementing ResNet101, VGG16, and Xception, modifying their layers for improved performance, and providing valuable insights into the effectiveness of different models in the given context.

Using Vision Transformers (ViTs) for diabetic retinopathy detection involves trade-offs between computational complexity and accuracy, resource usage and performance, and implementation complexity and scalability. A study conducted by Mehta and Rastegari (2021) introduced MobileViT, a lightweight, general-purpose vision transformer designed for mobile and resource-constrained environments. The MobileViT model innovatively combines standard convolutional neural networks (CNNs) with lightweight vision transformers, harnessing the strengths of both to form a compact yet powerful architecture. This hybrid structure incorporates mobile-specific optimizations such as depthwise separable convolutions and efficient self-attention mechanisms. To further enhance efficiency, techniques like model pruning, quantization, and parameter sharing are utilized to minimize model size and computational demands, while lightweight transformer blocks ensure the model's ability to capture long-range dependencies remains intact. MobileViT achieves competitive accuracy on benchmark datasets like ImageNet, maintaining a significantly smaller model size and lower latency compared to traditional ViTs and CNNs. The model excels in computational efficiency, evidenced by reduced inference times and lower memory usage. Despite the optimizations, MobileViT maintained a high accuracy rate of 90%, comparable to traditional ViTs and superior to many CNN models. In another study, Dosovitskiy et al. (2020) introduced Vision Transformers (ViTs) by leveraging the self-attention mechanism to capture

Table 12 Comparison of computationally efficient deep learning models

Str#	References	Algorithms used	Limitations
1	(Khan et al. 2023)	DR-CCTNet, which is a modified version of the Compact Convolutional Transformer (CCT) model	Less number of raw images are used in this study. The performance of the proposed model in real-time data is not assessed
2	(Kukkar et al. 2022)	Hybrid Genetic and Ant Colony Optimization (HGACO)	Did not explore using different angles of DR images, Also did not explore a patch-wise classification approach
3	(Raiaan et al. 2023)	Developed a base CNN model and optimized the base model to create the proposed RetNet-10model	A limited number of raw images were used. The performance of the proposed model on real-time data is not evaluated
4	(Nahiduz-zaman et al. 2023)	Contrast Limited Adaptive Histogram Equalization (CLAHE), Parallel Convolutional Neural Network (PCNN), and ExtremeLearning Machine (ELM)	Didn't compare their result with deep learning models. Comparison of results with CNN who is not using parallel convolution is missing
5	(Jian et al. 2023)	Convolutional neural networks (CNNs), ResNet as the backbone of the three cascade subnetworks; replaced some ordinary convolutions in ResNet with depthwise convolutions (DWConv)	The independent training of each subnetwork prevents feature fusion, which could improve the overall model performance
6	(Aujih et al. 2022)	Depth-wise separable convolution (DWSC) modules, EDR-Net blocks consisting of two DWSC modules with skip connections	Didn't evaluate the proposed design choice via ablation studies. Also, SEC module did not improve the DWSC-based DR-Net (EDR-Net)
7	(Aurangzeb et al. 2022)	ColonSegNet, a lightweight encoder-decoder architecture originally developed for colonoscopic image segmentation	The proposed model did not perform well on noisy images

global dependencies within retinal images. The study aimed to assess the performance of ViTs in comparison to traditional Convolutional Neural Networks (CNNs). Through optimization techniques such as layer-wise pruning and distillation, the inference time for the ViT was reduced by 30%, making it comparable to optimized CNNs.

Despite promising results, ViTs for DR detection require further research to validate performance across diverse datasets, enhance interpretability, and optimize efficiency for real-time applications, with the potential to revolutionize the diagnosis and management of DR.

3.11 Assessment of RQ4: what are the various approaches and methodologies of deep learning that are computationally efficient for DR detection?

In assessing the approaches and methodologies of deep learning for computationally efficient diabetic retinopathy (DR) detection, several innovative models and techniques were identified. The proposed EDR-Net (Aujih et al. 2022) in the literature addresses the computational inefficiency of the existing DR-Net architecture by introducing a depth-wise separable convolution module. This model achieves comparable predictive performance to state-of-the-art methods in detecting referable diabetic retinopathy (rDR) while significantly reducing computation costs, making it suitable for mobile device-based applications. Another notable approach (Kukkar et al. 2022) involves a Diabetic Retinopathy Classification (DRC) system, leveraging the Internet of Medical Things (IoMT), a deep learning model (RESnet), and a hybrid optimization (HGACO) algorithm. This system shows promise in early detection and classification of DR severity, presenting a potential solution for preventing blindness in diabetic patients. Additionally, the Symmetric Mask Pre-Training Vision Transformer (SMiT) introduces a transformer-based framework for grading pathological images (Zhang et al. 2023). By deviating from traditional convolutional neural network (CNN) models, SMiT achieves superior results in diagnosing colorectal cancer images, demonstrating the effectiveness of visual transformers in medical imaging applications. Furthermore, the Triple-DRNet (Jian et al. 2023) model adopts a triple-cascade network approach to effectively distinguish various DR lesions, enhancing grading performance with high accuracy on the APTOS 2019 Blindness Detection dataset. These approaches collectively showcase advancements in deep learning methodologies that prioritize computational efficiency for DR detection.

Khan et al. (2023) utilized the fine-tuned Compact Convolutional Transformer (CCT) model and introduced the DR-CCTNet model, a modification of the CCT model, for efficient training concerning computational complexity. In another study, Raiaan et al. (2023) proposed the RetNet-10 model for automatic DR detection. They merged three DR datasets (APTOS, Messidor2, IDRiD) to form a dataset of 5,819 images and applied image pre-processing, and data augmentation, and developed a shallow CNN base model. Finally, they experimented with optimized model components and hyperparameters to reduce the computational time. Aurangzeb et al. in their study employed a lightweight DL model based on ColonSegNet for retinal vessel segmentation with the aim to address the gap of reducing computational complexity in retinal vessel segmentation. The performance of the proposed model was assessed on three datasets (DRIVE, CHASEDB1, STARE) with superior results, making it suitable for low-end hardware devices (Aurangzeb et al. 2022). Another group of researchers introduced a contrast-limited Adaptive Histogram Equalization (CLAHE) pre-processing technique with a Parallel Convolutional Neural Network (PCNN) for fea-

ture extraction and an Extreme Learning Machine (ELM) for classification. Fundus images were preprocessed with CLAHE to enhance lesion visibility. They address the problem of reducing parameter complexity and layer complexity as compared to conventional CNN (Nahiduzzaman et al. 2023).

For future research, we suggest several avenues to enhance the performance and applicability of diabetic retinopathy detection models. Improving the performance of specific subnetworks, such as the NPDR-Net, and compressing network parameters will make models more suitable for real-world clinical applications. Lastly, systematic ablation studies to evaluate different design choices for architectures like EDR-Net, along with improvements to modules like the SEC module for better control of channel and spatial dimensions, will be instrumental in advancing the field. Furthermore, implementing different angles of diabetic retinopathy images and employing a patch-wise classification approach with fewer resources could enhance model accuracy and efficiency. Moreover, investigating the application of geometrical deep learning and graph neural networks may offer insights into the progression of diabetic retinopathy.

3.12 Assessment of RQ5: what are the key performance metrics and benchmark datasets commonly used to evaluate the effectiveness and efficiency of deep neural networks for DR detection?

An extensive examination was conducted on 85 carefully selected research papers to identify prevalent evaluation metrics employed in the detection of Diabetic Retinopathy (DR). The findings, as presented in Table 11, primarily focused on reporting the accuracy of each selected paper. However, it is noteworthy that the common evaluation metrics utilized across these studies encompass accuracy, specificity, sensitivity, and, in some papers, the Area Under the Receiver Operating Characteristic (AUC-ROC) curve. These metrics collectively contribute to a comprehensive understanding of the performance of DR detection models.

In addition to evaluating metrics, the study recognized the pivotal role played by datasets in the research process. A thorough analysis of the datasets employed in these studies was conducted, and the dataset characteristics are presented in detail in Table 13. This examination sheds light on the diverse attributes and features of the datasets utilized, providing valuable insights into the foundation upon which the DR detection models were developed and assessed in the respective studies.

3.12.1 Messidor-2

The Messidor-2 dataset serves as a valuable resource for advancing research in computer-assisted diagnoses of diabetic retinopathy. It comprises 1748 macula-centered eye fundus images, including 874 pairs from patients with diabetic retinopathy. The high-resolution PNG format images, each measuring 1440 x 960 pixels, have undergone expert grading to determine the severity of diabetic retinopathy. Additionally, the dataset facilitates the comparison and evaluation of various segmentation algorithms and image database management tools, offering annotations for select images to indicate the presence of lesions such as microaneurysms, hemorrhages, and exudates.

Table 13 Dataset characteristics

Dataset name	Total images	Image format	Image size	Accessible link
Messidor-2	1748	PNG	1440 x 960 pixels	https://www.adcis.net/en/third-party/messidor2/
IDRiD	516	JPEG	4288 x 2848 pixels	https://idrid.grand-challenge.org/
DDR	13,673	-	-	https://www.kaggle.com/datasets/mariaherrerot/ddrdataset
EyePACS	88,000	JPEG	-	https://www.kaggle.com/c/diabetic-retinopathy-detection/
DRIVE	40	JPEG (images), PNG (ground truth)	565 x 584 pixels	https://drive.grand-challenge.org/
CHASE_DB1	28	-	999 x 960 pixels	https://blogs.kingston.ac.uk/retinal/
STARE	20	JPEG	700 x 605 pixels	https://cecas.clemson.edu/~ahoover/stare/
DIARETDB0	130	JPEG	50-degree FOV	https://www.it.lut.fi/project/imageret/diaretdb0/
DIARETDB1	89	JPEG	1500 x 1151 pixels	https://www.it.lut.fi/project/imageret/diaretdb1/index.html
ORIGA	650	-	3072 x 2048 pixels	-
HRF	45	-	3304 x 2336 pixels	https://www5.cs.fau.de/research/data/fundus-images/
OCTID	500	JPEG	500 x 750 pixels	https://borealisdata.ca/dataverse/OCTID
APTOS 2019	3,600	-	-	https://www.kaggle.com/c/aptos2019-blindness-detection/

3.12.2 IDRiD

The Indian Diabetic Retinopathy Image Dataset (IDRiD) is a comprehensive resource focused on diabetic retinopathy (DR) and diabetic macular edema (DME), tailored to represent the Indian population. With 516 high-resolution fundus images at 4288 x 2848 pixels and a 50-degree field of view, the dataset includes pixel-level annotations of DR lesions and normal retinal structures, aligning with international standards for disease severity grading. Notably, IDRiD stands out as the sole dataset offering such detailed annotations for an Indian demographic. Captured in authentic clinical settings, it ensures practical applicability

and clinical relevance. The dataset's JPEG format for images and XML format for annotations, along with challenges such as image variability and potential biases, underline its significance in population-specific research.

3.12.3 DDR

The DDR dataset is specifically crafted to support research and development in the realm of diabetic retinopathy (DR) by facilitating detection, classification, and lesion segmentation. Comprising 13,673 fundus images sourced from 147 hospitals spanning 23 provinces in China, the dataset ensures geographic diversity and excludes poor-quality images through meticulous quality control. This dataset proves instrumental for advancing DR-related research, offering a substantial resource for algorithm development and evaluation.

3.12.4 EyePACS

The EyePACS dataset serves a multifaceted purpose, aiming to advance early detection of diabetic retinopathy (DR) and other eye diseases through telemedicine and digital imaging. The dataset comprises over 88,000 high-resolution digital retinal images captured using diverse cameras and imaging protocols. Each image undergoes grading for DR presence and severity by trained human graders, adhering to a standardized protocol. The dataset includes limited patient metadata such as age, gender, and primary care provider information. The images are stored in JPEG format, while grading information is in XML format.

3.12.5 DRIVE

The dataset consists of retinal fundus images captured using a Canon CR5 non-mydratiac 3CCD camera with a 45° field of view, encompassing 40 images, each accompanied by two manually segmented ground truths for blood vessels. The dataset, available in color JPEG format for images and PNG format for ground truth vessel segmentations, bears certain considerations such as its relatively small size, limited diversity in image acquisition devices and patient populations, and the potential for bias in ground truth annotations.

3.12.6 CHASE_DB1

The CHASE_DB1 dataset serves the primary purpose of advancing retinal vessel segmentation algorithms, providing a valuable resource for development and evaluation in this domain. Comprising 28 color fundus images of retinas with a resolution of 999 x 960 pixels, the dataset features images captured from both eyes of 14 children (7 female, 7 male). Derived from the Child Heart and Health Study in England (CHASE), the images were obtained using a Nidek NM-200-D fundus camera with a 30-degree field of view. Despite its relatively small size, which may pose challenges for training complex models, the dataset's focus on children's retinas provides a unique perspective, though its generalizability to adult populations should be considered.

3.12.7 STARE

The dataset, primarily geared towards retinal vessel segmentation, encompasses 20 color fundus images with a resolution of 700 x 605 pixels. It provides two sets of ground-truth vessel annotations: Set 1 by Adam Hoover, commonly employed for training and testing, and Set 2 by Valentina Kouznetsova, serving as a human baseline. While the project comprises over 400 raw images, only 20 are publicly available with annotations. Widely utilized in research studies, the dataset is marked by its relatively small size, and limited diversity of images in terms of ethnicity and disease types. Accessible via the STARE project website,

3.12.8 DIARETDB0

The DIARETDB0 dataset is designed for the development and evaluation of algorithms aimed at automated diabetic retinopathy (DR) detection. Comprising 130 color fundus images captured with 50-degree field-of-view digital fundus cameras, the dataset includes 20 normal images and 110 images exhibiting signs of DR. The dataset, labeled as "calibration level 0," suggests potential variability in image quality.

3.12.9 DIARETDB1

The DIARETDB1 dataset serves as a resource for diabetic retinopathy detection and assessment, comprising 89 color fundus images with a resolution of 1500 x 1151 pixels in JPEG format. With a focus on aiding research and algorithm development for diabetic retinopathy, the dataset encompasses images from 89 patients with varying degrees of the condition. It is relatively small size and limited diversity in image characteristics pose challenges to the generalizability of algorithms trained on it.

3.12.10 ORIGA

The ORIGA dataset, standing for Online Retinal Fundus Image Dataset for Glaucoma Analysis and Research, is dedicated to glaucoma detection and analysis. Comprising 650 retinal fundus images with a resolution of 3072 x 2048 pixels, the dataset is distributed between 482 healthy images and 168 glaucomatous images. Annotations include class labels distinguishing healthy from glaucoma cases, as well as detailed contours of the optic disc (OD) and optic cup (OC), along with cup-to-disc ratio (CDR) values.

3.12.11 HRF

The HRF dataset, designed for retinal vessel segmentation and image quality assessment, features high-resolution fundus images with a resolution of 3304 x 2336 pixels. Comprising a segmentation dataset of 45 images and an image quality assessment dataset of 18 image pairs, the dataset includes conditions such as healthy, diabetic retinopathy, and glaucoma. Annotations consist of binary gold standard vessel segmentation masks and field of view (FOV) masks. Organized into 15 subsets, each containing healthy, diabetic retinopathy, and glaucoma images, the dataset is divided into 22 training images and 23 testing images for the segmentation dataset. The image quality assessment dataset comprises 18 pairs of

images capturing the same eye under good and poor quality conditions, utilizing a Canon CR-1 fundus camera.

3.12.12 OCTID

The Optical Coherence Tomography Image Retinal Database (OCTID) is an open-source dataset designed for ophthalmology research, comprising over 500 high-resolution retinal images captured using Optical Coherence Tomography (OCT) technology. These images are systematically categorized into five classes, including Normal, Macular Hole, Age-related Macular Degeneration, Central Serous Retinopathy, and Diabetic Retinopathy, encompassing different disease stages. The dataset presents images in JPEG format resized to 500x750 pixels.

3.12.13 APTOS 2019

APTOS 2019, hosted by the Asia Pacific Tele-Ophthalmology Society, held significant importance in 2019 by concentrating on addressing the challenge of diabetic retinopathy (DR), a leading cause of blindness. A pivotal contribution was the release of the APTOS 2019 dataset, featuring over 3,600 fundus images from rural India, expertly classified by DR severity (Albadr et al. 2022).

4 Discussion and future directions

This section provides an overview and analysis of the outcomes pertaining to the systematic literature review.

Integrating advanced computational models like Vision Transformers (ViTs) into clinical practices offers significant potential for enhancing patient outcomes but also presents several challenges that must be addressed for successful implementation in real-world healthcare settings (Parvaiz et al. 2023). These challenges include system integration and interoperability with existing healthcare information systems, ensuring data privacy and security in compliance with regulations, and overcoming the computational resource limitations that may exist in smaller or resource-limited healthcare environments. Additionally, clinical acceptance can be hindered by workflow disruptions and skepticism towards AI-driven decision-making processes, while extensive training and ongoing support for healthcare professionals are necessary to effectively utilize these technologies. Despite the hurdles and difficulties, There are enough benefits, e.g. patient outcomes can be improved significantly using more accurate and timely diagnoses, operational efficiency can be increased by automating the daily tasks, insights can be driven from data to help in clinical decision-making, and the scalability of advanced diagnostic tools to reach underserved areas. Addressing the challenges through careful planning and ensuring model transparency will be key to unlocking the full potential of these technologies in healthcare.

Vision Transformers (ViTs) are helpful in clinical sites for the detection of diabetic retinopathy and their implementation requires cautious consideration of computational load and hardware necessities. For some ViTs like ViT-B/16 requires significant computational sources because it has almost 86 million parameters and a self-attention mechanism requir-

ing $O(n^2 \cdot d)$ operations. For example, when a 224x224 pixel image is processed using ViT-B/16 requires approximately 1.0 s on GPUs like the NVIDIA RTX 3090. These models can be trained using high-performance GPUs such as NVIDIA A100 with substantial memory requirement of may be more than 20 GB and energy requirement of almost 400 watts per GPU. The deployment of ViTs can become challenging in clinical settings due to the budget limitations and lack of advanced infrastructure in these environments. Therefore, ViTs can be optimized in these clinical environments using techniques such as pruning and quantization. The cost and performance of these ViTs can also be balanced by means of mid-range GPUs or edge devices. In future, the focus should be on the development of proficient ViT structural design or hybrid devices with moderate computational requirements and high accuracy, thus improving the practicality of using these devices in clinical applications.

To ensure the generalizability of computational models for diabetic retinopathy detection, it's crucial to think and analyze how these models perform and might be translated in various regions, especially considering the available quality of healthcare and how common diabetes is in different regions. The prevalence of diabetes rates can vary a lot on the basis of different factors e.g. lifestyle, genetics, and socioeconomic conditions. In the regions and areas where diabetes is more common (Shera et al. 2007), these models will be needed to consider and handle a big range of disease severity. On the other side, in areas where diabetes patients are less common, the models' accuracy and reliability can be compromised as models will have less data to learn. The local healthcare system also matters a lot. In developed countries, advanced models can be easily integrated into existing healthcare systems as their healthcare systems are already technology-oriented. However in countries like Pakistan, it will be difficult to integrate these models as they have limited access to high-performance computing and staff is less trained to adopt the technology. Therefore, adapting the models to work in different environments, such as by making them less demanding on technology or creating simpler interfaces, is crucial for ensuring that these technologies can benefit people worldwide. This is essential for making sure the findings are useful and effective across different populations and healthcare systems.

4.1 Thematic analysis

This systematic literature review aimed to assess contemporary lightweight models for the detection of Diabetic Retinopathy. To fulfil this objective, we constructed a taxonomy hierarchy by thoroughly examining and analyzing the chosen studies, as illustrated in Fig. 6. The investigation delved into challenges and advancements across various dimensions, encompassing algorithms and models for DR detection, computationally efficient models, and image processing and enhancement techniques. Furthermore, these dimensions were subdivided into smaller domains to reveal the depth of each aspect contributing to the enhancement of model performance in Diabetic Retinopathy detection.

Moreover, a thematic analysis is conducted to extract relevant relationships from the chosen studies, and subsequently, these relationships are coded (Thomas and Harden 2008; Senapati et al. 2024; Schloemer and Schröder-Bäck 2018). The codes are derived from the existing literature review presented in Table 14. Initially, papers employing diverse algorithms and models are selected and coded as VT, DL, TL, and ML. Subsequently, the papers are categorized based on computational complexity, severity classification, and are assigned codes such as LW, OM, CS, SE, and GS. Lastly, papers are selected based on image process-

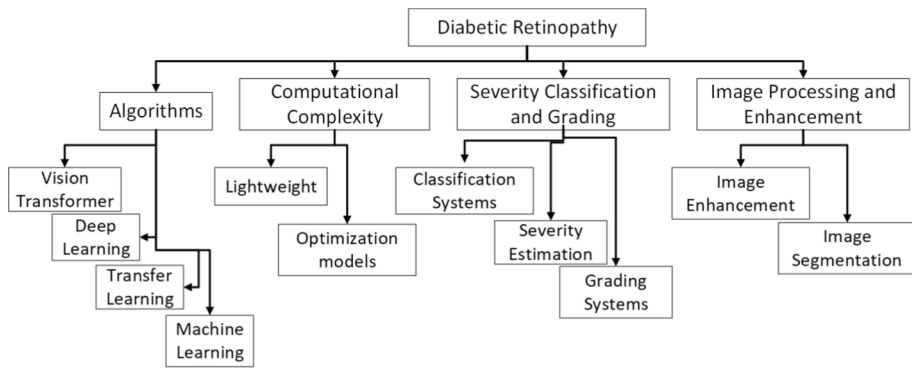


Fig. 6 Taxonomy of diabetic retinopathy

ing and image enhancement techniques, with codes assigned as IE and IS. The selected studies are thoroughly examined by evaluating and analyzing their objectives, methodologies, areas of discussion, and limitations.

4.2 Future recommendations

Researchers focusing on advancing diabetic retinopathy (DR) detection should immediately prioritize integrating multimodal data for comprehensive early diagnosis systems, developing sophisticated deep learning architectures for multimodal data fusion, and conducting rigorous validation through cross-validation and longitudinal studies to assess both accuracy and long-term treatment outcomes (Bilal et al. 2021, 2022, 2024a, 2021). Concurrently, they should explore semi-supervised learning approaches to mitigate the challenge of limited labeled datasets by leveraging techniques such as self-training and data augmentation to enhance model performance while reducing dependency on expert-labeled data. Evaluating these models against traditional supervised methods and ensuring robustness across diverse datasets will be crucial, with emerging technologies such as federated learning, explainable AI, and using lightweight models or lightweight vision transformers offering transformative potential to accelerate progress in this critical area of medical research (Bilal et al. 2024b, 2024; Sadiq et al. 2021; Hasan et al. 2024).

To improve the efficiency and accuracy of diagnosis systems for DR detection, this study discovered few problems that can be addresses in future.

4.2.1 Problem 1: integrating iultimodal iata for comprehensive early diagnosis systems:

It is observed in the literature that traditional studies, which exclusively rely on imaging data for Diabetic Retinopathy (DR) diagnosis, may overlook crucial factors influencing the development of the disease. An enhanced understanding of a patient's health profile, including clinical data such as age, blood sugar, and blood pressure, allows for more personalized and nuanced treatment decisions (Kamarudin et al. 2022; Wenhua et al. 2024). However, the majority of current studies focused on early DR diagnosis fail to integrate multimodal data. This research gap limits the comprehensive understanding of patient health, poten-

Table 14 Taxonomy coding scheme for SLR

Domain	Code	Sub domain	References
Algorithms	VT	Vision Transformer	Khan et al. (2023), Dutta et al. (2023), Hossain et al. (2023), Dihin et al. (2022), Li et al. (2023), Zhang et al. (2023), Wang et al. (2023)
	DL	Deep Learning	Luo et al. (2023), Ali et al. (2023), Raiaan et al. (2023), Beham and Thanikaiselvan (2023), Krishnamoorthy et al. (2023), Jena et al. (2023), Sau and Bansal (2022), Bilal et al. (2022), Bilal et al. (2022), Özbay (2023), Aurangzeb et al. (2022), Dayana and Emmanuel (2022), Elaouaber et al. (2023), Aziz et al. (2023),
	TL	Transfer Learning	Bilal et al. (2022), Al-Smadi et al. (2021), Bhardwaj et al. (2021)
	ML	Machine Learning	Rayavel and Murukesh (2022), Bansode et al. (2023), Pugal Priya et al. (2022), Nahiduzzaman et al. (2023), Gundluru et al. (2022), Ragab et al. (2022), Kshirsagar et al. (2022), Jian et al. (2023), Nguyen et al. (2021)
	LW	Lightweight	Raiaan et al. (2023), Khan et al. (2023), Aurangzeb et al. (2022), Nahiduzzaman et al. (2023), Aujih et al. (2022), Ragab et al. (2022), Zhang et al. (2023), Wang et al. (2023), AbdelMaksoud et al. (2022), Abirami and Kavitha (2023), Wan et al. (2023)
Computational Complexity	OM	Optimization Models	Dayana and Emmanuel (2022), Özbay (2023)
	CS	Classification Systems	AbdelMaksoud et al. (2022), Qureshi et al. (2021), Goel et al. (2021)
	SE	Severity Estimation	Luo et al. (2023), Bodapati et al. (2021), Das et al. (2021)
	GS	Grading Systems	(Sau and Bansal 2022; Zhang et al. 2021; Gunasekaran et al. 2022), (Aurangzeb et al. 2022)
Image Processing and Enhancement	IE	Image Enhancement	Abbood et al. (2022), Naik et al. (2023), Dayana and Emmanuel (2022)
	IS	Image Segmentation	(Aujih et al. 2022; Khaparde et al. 2023; Liu et al. 2022)

tially hindering the accuracy and effectiveness of treatment decisions. The incorporation of data from diverse modalities can play a vital role to establish early diagnosis systems that account for the broader health context of individuals. Standard diagnostic metrics like accuracy, sensitivity, and specificity can be used to evaluate the effectiveness of the multimodal data integration strategy. The performance of the multimodal approach can be compared with traditional unimodal methods that solely rely on imaging data. This involves examining the additional value and accuracy improvement attained by integrating patient-specific information. We can conduct cross-validation on diverse patient populations to assess the generalizability of the multimodal approach. It can also be assessed that how well the model

performs across different demographic groups, ensuring the robustness and reliability of data integration across various patient profiles. Furthermore, a longitudinal study can be conducted to analyze treatment outcomes based on the comprehensive early diagnosis system. By addressing this research problem and employing the specified evaluation methods, researchers can advance the field of Diabetic Retinopathy diagnosis, creating more inclusive and effective early diagnosis systems. This approach holds the potential to significantly impact patient care and outcomes in the context of Diabetic Retinopathy.

4.2.2 Problem 2: semi-supervised deep learning based model for detection of DR

It has been noted in the existing literature and within selected papers that, despite the existence of public datasets designed for Diabetic Retinopathy (DR) detection, the size of the training and testing image sets within these datasets is limited. The primary challenge is the need for a large number of labeled fundus images and the associated cost of acquiring expert services for this purpose. Hence, there is a need to investigate strategies and methodologies to reduce dependency on expert services for image labeling. This problem is important as it addresses a key bottleneck in the development of supervised deep-learning models for Diabetic Retinopathy detection, making the process more cost-effective and scalable. A subsequent investigation within this domain could be undertaken, focusing on the development of a semi-supervised deep learning model designed for the detection of Diabetic Retinopathy (DR) using minimal labeled data. This is crucial as it offers a potential solution to the scarcity of labeled fundus images, enabling the model to leverage both labeled and unlabeled data for effective learning. The effectiveness of the proposed strategies in reducing dependency on expert services and leveraging limited labeled data can be quantitatively evaluated through metrics such as classification accuracy, precision, recall, and F1-score. This provides a robust assessment of the models' performance in DR detection. To gauge the impact of the proposed semi-supervised learning approaches, a comparative analysis with traditional supervised methods should be conducted. This involves evaluating the performance of models trained with limited labeled data against models trained with a large fully labeled dataset. This comparison will help to establish the efficacy of the proposed methods. However, it is essential to assess how well the models generalize to unseen data and variations in image characteristics. Generalization and robustness testing involve evaluating model performance on diverse datasets, including those with different resolutions, qualities, and sources. This provides insights into the models' adaptability to real-world scenarios. Moreover, considering the aim of reducing costs associated with expert services, evaluation methods should include metrics related to resource utilization. This involves assessing the computational efficiency, time, and cost savings achieved through the proposed semi-supervised learning methods compared to traditional approaches. Solving this research problem and applying the defined evaluation methods will contribute to the development of more cost-effective and scalable solutions for Diabetic Retinopathy detection, addressing the challenges associated with the need for a large number of labeled fundus images.

5 Conclusion

This SLR has been conducted to develop a comprehensive understanding of computationally efficient deep-learning models for the detection of diabetic retinopathy (DR). The study adopts a systematic literature review approach to ensure a thorough exploration of challenges and their corresponding solutions. A meticulous search query, comprising a cluster of keywords pertinent to DR detection, is crafted, and the ensuing results are systematically assessed. To refine the scope, proper inclusion–exclusion criteria and rigorous quality assessments are applied, leading to the identification and extraction of 85 relevant articles. The time frame for this extensive search spans from January 2021 to September 2023. This literature review distinguishes itself by not only focusing on prevalent deep learning methods for DR detection but also by incorporating a novel emphasis on the computational aspect of these models. The primary objective is to understand and evaluate computational efficiency, especially pertaining to real-time detection of DR. Seven major digital repositories have been meticulously explored for this search, and papers published in reputable journals only are included in this SLR. In the future, multi-modal data will be integrated for comprehensive diagnosis of DR detection. Moreover, semi-supervised deep learning models will be introduced to learn from minimal labeled data.

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Data availability statement Since this is a systematic literature review paper so the data used in the paper to report results are explained in section 3.3. The relevant queries and search strategy we used to extract relevant papers from digital libraries are given in table 6.

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